Vendor Performance Analysis

SQL + Python + Power BI | 119 vendors | $441M in sales

VICKY JEERU
DATA ANALYST | POWER BI DEVELOPER
Hyderabad, India
jeeruvicky51@gmail.com
+91 91541 39298
LinkedIn **Portfolio** **GitHub**

$441M	30.4%	65.7%	$2.7M
total sales on $307M of purchases	overall gross margin ($134M gross profit)	of purchase dollars come from just 10 of 119 vendors	net capital tied up in unsold inventory ($9.6M gross overstock)

Executive Summary

Retail and wholesale margins depend on buying from the right vendors, at the right price, in the right quantities. Using **SQL, Python (Pandas) and Power BI**, I built a vendor summary from four raw tables and analyzed **8,565 vendor-brand records across 119 vendors** to find where profit is made, where cash is trapped, and where the business depends too heavily on a few suppliers.

> **Bottom line:** the business is **concentrated and thin-margined where it matters most**. The top 10 vendors supply 65.7% of purchases, and the three biggest-buying vendors with the weakest margins (Diageo, Pernod Ricard, Jim Beam at 25-26% vs the 30.4% average) also hold the most unsold stock. The "bulk discount" in the data turns out to be a product-mix effect, not a real volume discount, so **negotiating genuine volume pricing is the largest untapped lever**.

Key Findings at a Glance

- **Vendor concentration:** top 3 vendors = 32.4% of purchases (Diageo alone 16.3%), while the top 10 brands are only 13.0% of sales, so the risk sits at vendor level.
- **Bulk purchasing:** large orders pay 72% less per unit, but cost is about 66% of list price in every size band, so there is no volume discount yet.
- **Sales are skewed:** the top 25% of records generate 88.1% of sales and 85.7% of gross profit, at a 29.5% margin.
- **Low-sales, high-margin brands:** 198 brands average a 79.8% margin but total only $55.9K in sales (0.01%).
- **Inventory:** $2.71M of net unsold capital, led by Diageo ($722K), Jim Beam ($555K) and Pernod Ricard ($471K).

Business Problem

The company must avoid losses from inefficient pricing, slow inventory turnover and vendor dependency. The analysis asks: **which brands need pricing changes, which vendors drive profit, whether bulk buying lowers unit cost, and how much capital sits in unsold stock.**

Tools & Techniques

Area	What I used
SQL	SQLite: CTEs, multi-table joins and LEFT JOINs to build `vendor_sales_summary` from four tables
Python	Pandas, NumPy, Seaborn, Matplotlib; SciPy for confidence intervals and hypothesis tests
Power BI	One-page dashboard: KPI cards, donut, bar, funnel and scatter visuals

1. Data Pipeline & Preparation

The analysis starts from four raw tables. A SQL script combines them into one analysis-ready table, and Python then cleans it and adds the profitability metrics.

Stage	What it does	Why it matters
1. FreightSummary (CTE)	Totals freight cost per vendor from `vendor_invoice`	Brings logistics cost into the vendor view
2. PurchaseSummary (CTE)	Joins `purchases` to `purchase_prices` on Brand (purchase price > 0) and sums quantity and dollars per vendor-brand	Gives cost and list price per brand
3. SalesSummary (CTE)	Sums sales quantity, dollars, price and excise tax per vendor-brand from `sales`	Gives revenue per brand
4. LEFT JOINs	Keeps every purchased brand, even if it never sold, then orders by purchase dollars	Unsold stock stays visible instead of disappearing
5. Python cleaning	Casts types, fills nulls with 0, trims text, adds GrossProfit, ProfitMargin, StockTurnover, SalesToPurchaseRatio	Creates the metrics used in every finding

What the raw summary table revealed (10,692 records)

Issue found	Detail	Handling
Loss-making records	Gross profit as low as -$52,002.78	Excluded (GrossProfit > 0)
Undefined margins	Profit margin of -infinity where sales were $0	Excluded (ProfitMargin > 0)
Never-sold stock	Records with zero sales quantity	Excluded (SalesQuantity > 0)
Extreme values	Purchase price up to $5,681.81 vs a $24.39 mean; stock turnover from 0 to 274.5	Kept: premium products and fast movers are real
Freight	$0.09 to $257,032.07, but it is a vendor-level total repeated on every brand row	Not used in profit; summing it would overcount

Result: 10,692 raw records reduced to **8,565 analysis records (119 vendors, 8,545 brand codes)**. Records that were loss-making or unsold are excluded, which is stated again under limitations.

Metric definitions

Metric	Formula
Gross profit	Total sales dollars - total purchase dollars
Profit margin (%)	Gross profit / total sales dollars x 100
Stock turnover	Sales quantity / purchase quantity (below 1 means stock left unsold)
Unsold capital	(Purchase quantity - sales quantity) x purchase price
Unit purchase price	Purchase dollars / purchase quantity

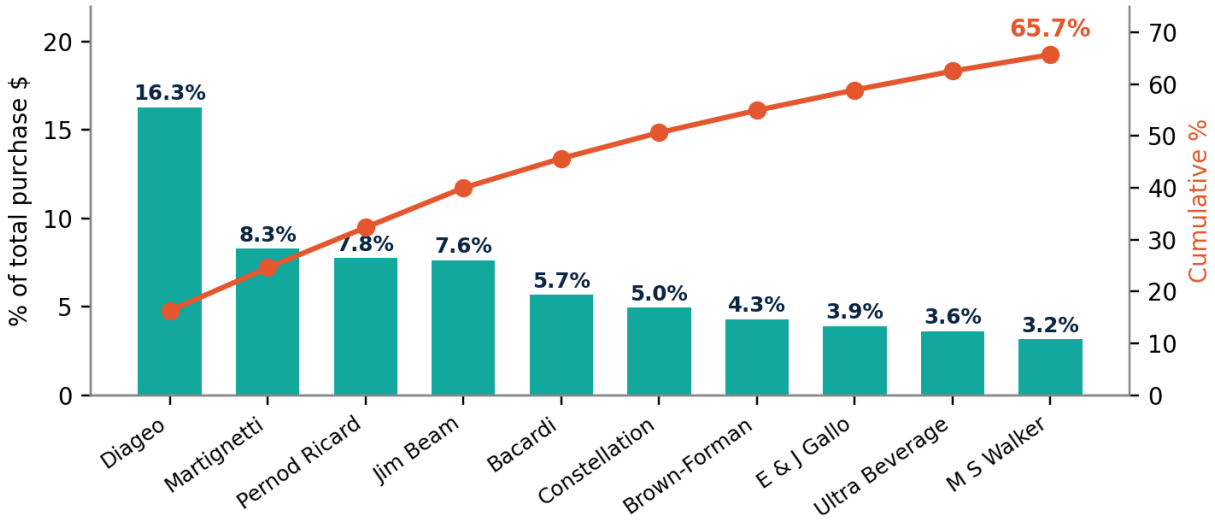
Correlation check: purchase price has no meaningful link to sales (-0.01) or gross profit (-0.02), purchase and sales quantity move almost together (0.999), and stock turnover correlates 0.40 with margin. That last link is partly mechanical: records with turnover above 1 sold older stock that is not in this period's purchase cost.

Vicky Jeeru | Data Analyst | jeeruvicky51@gmail.com | +91 91541 39298

2. Vendor Concentration & Profitability

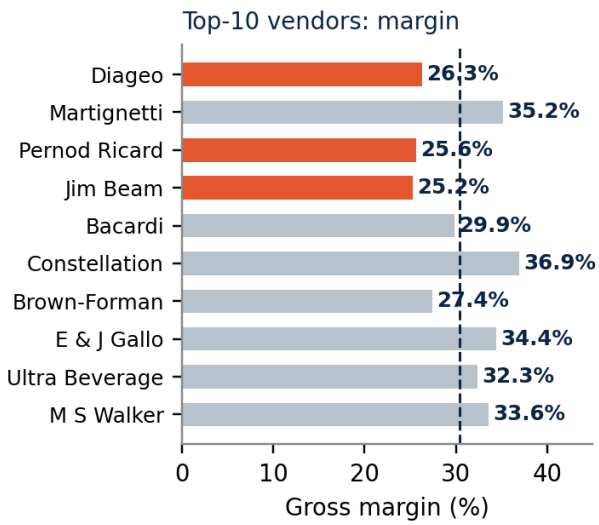
Finding 1: Purchasing depends heavily on a few vendors

The top 10 vendors account for **65.7%** of purchase dollars, and the other 109 vendors share the remaining 34.3%. Diageo alone is 16.3%, and the top 3 are 32.4%.



Business insight: a supply disruption or price increase at one of the top vendors would hit a large share of inventory. Sales follow the same pattern (top 10 vendors = 65.2% of sales), so the dependency runs both ways.

Finding 2: The biggest vendors are not the most profitable



Dashed line = overall margin of 30.4%. Weighted by sales, the top 10 vendors earn **29.8%** versus **31.4%** for all other vendors.

- **Diageo (26.3%), Pernod Ricard (25.6%) and Jim Beam (25.2%)** are three of the top four by purchases yet sit 4-5 points below the average margin
- **Constellation (36.9%) and Martignetti (35.2%)** earn the best margins among the large vendors
- The same three low-margin vendors also hold the most unsold stock (see section 4)

These are the vendors where negotiating better cost terms would pay off most.

Top brands by sales

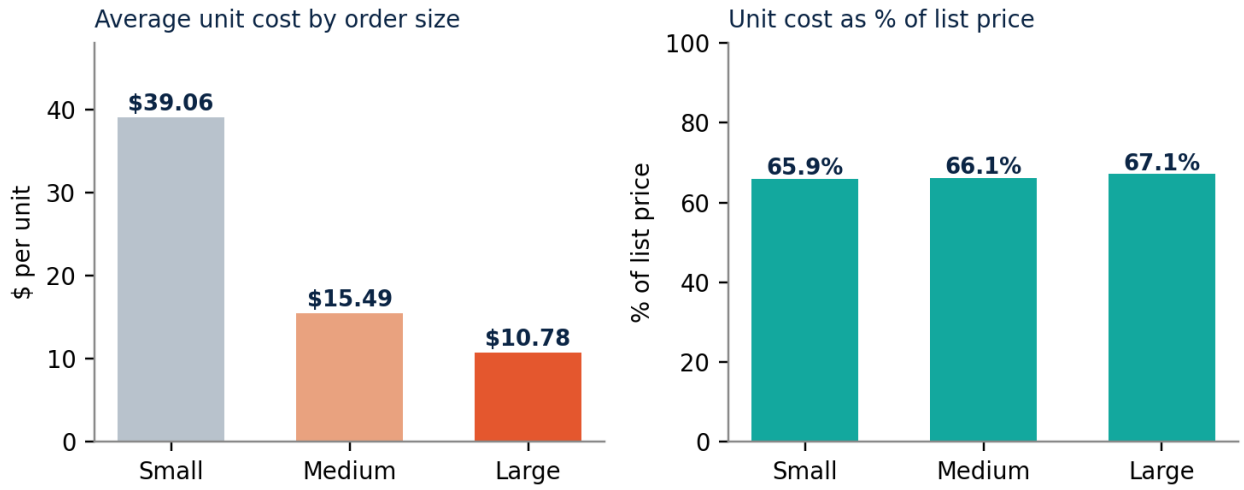
Brand	Sales	Gross margin
Jack Daniels No 7 Black	$7.96M	26.9%
Tito's Handmade Vodka	$7.40M	24.1%
Grey Goose Vodka	$7.21M	27.0%
Capt Morgan Spiced Rum	$6.36M	27.3%
Absolut 80 Proof	$6.24M	25.6%

The ten best-selling brands together contribute only 13.0% of sales ($57.2M), so brand risk is spread out while vendor risk is not.

3. Pricing & Product Opportunities

Finding 3: The bulk-purchase discount is really a product-mix effect

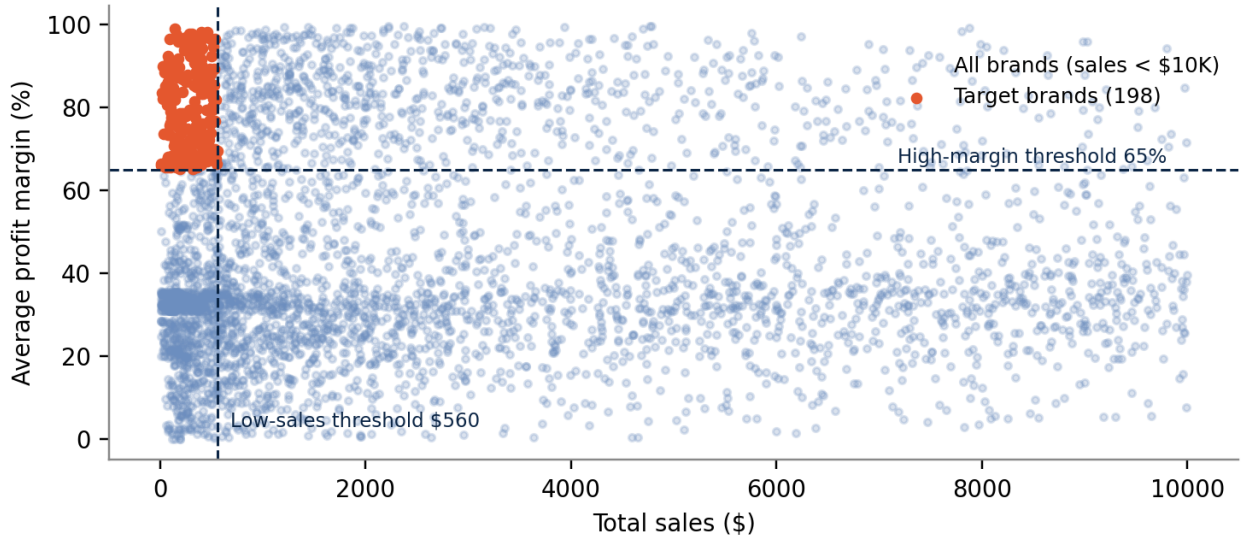
Splitting records into three equal groups by purchase quantity shows large orders paying far less per unit. That looks like a volume discount, but comparing each unit cost with the brand's **list price** tells a different story.



Business insight: unit cost is **65.9%, 66.1% and 67.1% of list price** for small, medium and large orders. The 72% gap in dollars per unit arises because large orders are mostly cheaper products (average list price $15.89 vs $57.96 for small orders). Within the same brand, order size and unit price are essentially uncorrelated (about 0.02 across 20 repeated brands). **There is no volume discount in the data yet, which makes negotiating one a clear opportunity**, since large-order records generate 90.8% of sales.

Finding 4: Low-sales, high-margin brands

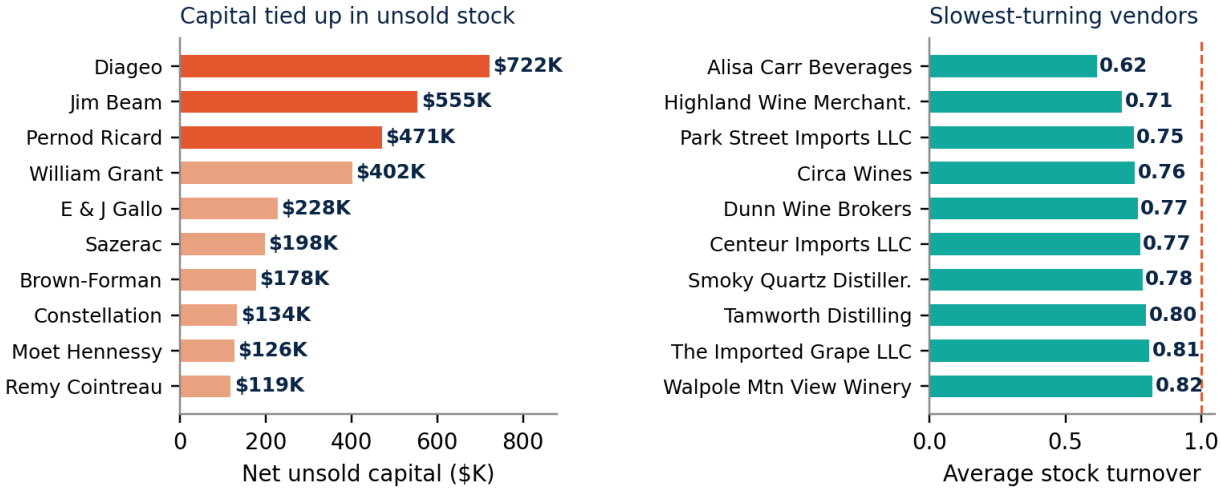
Brands in the bottom 15% of sales but the top 15% of average margin (above 65%) are candidates for promotion or repricing.



- **198 brands** qualify, with an average margin of **79.8%**, but together they sold only **$55.9K** (0.01% of sales; median brand $288).
- Because the base is so small, even a large percentage lift in volume adds little revenue, so pilot promotions on a few brands first and track the result before scaling.

4. Inventory Efficiency & Statistical Validation

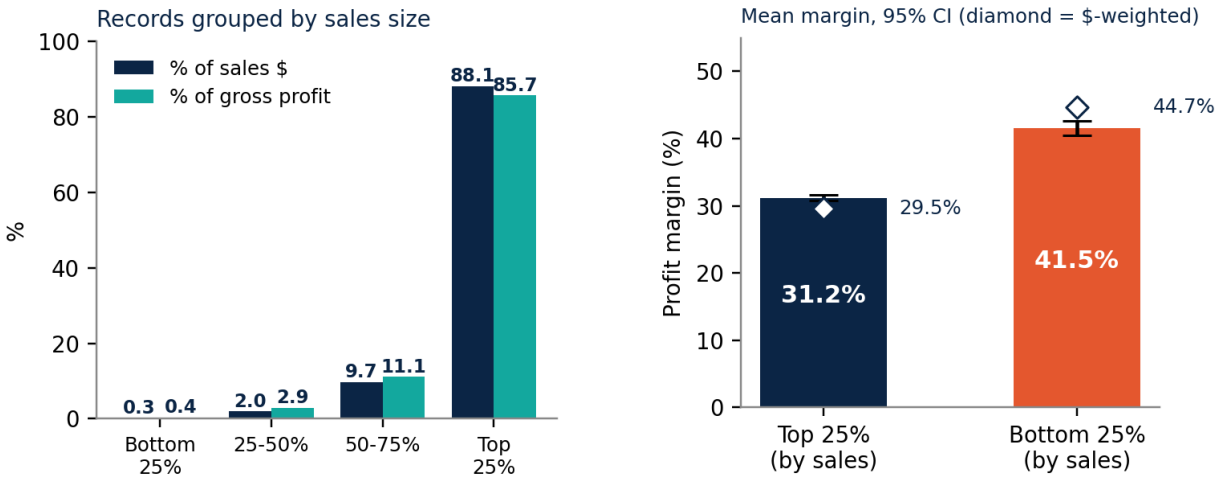
Finding 5: Slow-moving stock and locked capital



- Net unsold capital is **$2.71M**. Behind that net figure sit **4,229 overstocked records worth $9.55M**, offset by 3,485 records that sold more than they bought ($6.85M, sold from older stock).
- **Diageo ($722K), Jim Beam ($555K), Pernod Ricard ($471K) and William Grant ($402K)** hold the most net unsold capital; 107 vendors have slow-moving records (turnover below 1), the slowest being Alisa Carr Beverages (0.62) and Highland Wine Merchants (0.71).

Finding 6: Margins differ between high- and low-selling records

I compared the top 25% of records by sales (over $40.6K each) with the bottom 25% (under $1.4K each), 2,142 records per group, using 95% confidence intervals and a hypothesis test.



Group	Average margin	95% confidence interval	Sales-weighted margin
Top 25% by sales	31.2%	30.7% to 31.6%	29.5%
Bottom 25% by sales	41.5%	40.5% to 42.6%	44.7%

Hypothesis test: H0 says there is no difference in mean margin between the groups. Welch's t-test gives t = -17.6, **p < 0.001** (a rank-based Mann-Whitney test agrees), so H0 is rejected. Small records do carry higher margins, **but they are 0.25% of sales and 0.37% of gross profit**, while the top 25% deliver 88.1% of sales and 85.7% of gross profit. Cost and pricing improvements on large records matter far more than margin on small ones.

5. Dashboard, Recommendations & Next Steps

Power BI dashboard

The results are presented in a one-page interactive report (12 visuals) built on the cleaned summary plus three helper tables for brand performance, purchase contribution and low-turnover vendors.

Component	Purpose
6 KPI cards	Purchases, sales, gross profit, margin, unsold capital and purchase contribution
Donut and 2 bar charts	Purchase contribution by vendor, top vendors by sales, top brands by sales
Funnel and scatter	Low-turnover vendors and low-sales, high-margin brands (target flag)

Recommendations

#	Action	Evidence
1	**Negotiate real volume discounts** with the largest vendors; tie order size to price tiers	Cost is about 66% of list price at every order size; large orders = 90.8% of sales
2	**Reduce dependency** on the top vendors by qualifying alternates, and renegotiate costs with Diageo, Pernod Ricard and Jim Beam	Top 10 = 65.7% of purchases; those three earn 25-26% margins vs 30.4% average
3	**Clear slow stock:** cut reorder quantities and run clearance on overstocked vendor lines	$2.71M net ($9.55M gross) unsold; Diageo, Jim Beam, Pernod Ricard, William Grant lead
4	**Pilot promotions on a few of the 198 high-margin, low-sales brands** and measure the lift before scaling	Avg margin 79.8% but only $55.9K total sales
5	**Protect margin on the top-sales quartile**, where each point of margin moves the most profit	Top 25% = 88.1% of sales and 85.7% of gross profit at 29.5% margin

Limitations

- Loss-making and unsold records (2,127 of 10,692, or 19.9%) were excluded, so profit figures describe profitable records only and overstate overall profitability.
- Gross profit here is sales minus purchases in the period (a cash view), not cost of goods sold, so stock bought but not yet sold lowers profit.
- Freight is a vendor-level total that is not allocated to brands, and the data has no dates, so trends over time are not studied.

Next steps

- Analyze the 2,127 excluded records separately to quantify losses and dead stock.
- Allocate freight to brands and use sales-weighted margin (total gross profit / total sales) as the dashboard KPI.
- Add time (monthly purchases and sales) to track turnover and seasonality.

Vicky Jeeru | Data Analyst & Power BI Developer | Hyderabad, India
jeeruvicky51@gmail.com | +91 91541 39298 | GitHub | Portfolio | LinkedIn
Code and dashboard: https://github.com/jeeruvicky51-bit/Vendor-Performance-Analysis